

Abdul Khurram

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Available for Co-ops/Internships (Open to Relocation): May–Dec 2026 | Full-time: May 2027+

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Apache Spark, Scala, Airflow, Kafka, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered secure **Scala** pipelines orchestrated by **Airflow DAGs** to process **50M+ daily transactions** from **Delta Lake**, enforcing strict **RBAC/encryption** controls on **petabyte-scale sensitive** financial data.
- Optimized **Spark** jobs by refactoring **SQL** queries and tuning **AQE/cluster memory**, solving resource bottlenecks and reducing **latency** by **40%**.
- Developing **multi-region** data pipelines with built-in **fault tolerance**, **idempotent** re-runs, and automated **failover** for critical banking workloads.

Software Engineer Intern | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Engineered **distributed async pipelines** and real-time **ETL** on **AWS (SQS/S3)** with secure **Flask/FastAPI** backends, processing **thousands of posts/day** with stream-based analytics and **fault-tolerant** automation.
- Raised scheduler reliability to **99.9%** by integrating **Playwright E2E** gates into **GitHub Actions CI/CD** with **Docker** containers, adding **retry/backoff** and **hardening auth/session**.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led development of an **AI-driven** VR clinical simulation in **Unreal (C++)**, collaborating with UBC's School of Nursing to train nurse practitioners.
- Designed a novel **multithreaded supervisory architecture (C++/Python)** that monitors AI patient behavior and enforces clinical protocol adherence.
- Evaluated system performance, achieving **40%** latency reduction and **90%** inference cost savings through pipeline optimizations.

PROJECTS

Backer | *TypeScript/Next.js, AWS (DynamoDB, S3, SES), NextAuth*

Hack The Coast 2026

- Architected a **DynamoDB**-backed data layer (**10+ tables**) with typed CRUD, batch operations, and **transactional writes** powering an investor feed, messaging, and investment workflows.
- Trained an **XGBoost ranker** scoring startups on 6 features (stage affinity, text similarity, tag overlap, popularity) from **DynamoDB** event streams.
- Integrated **S3** uploads, **SES** emails, and **NextAuth** middleware with **role-based protection**.

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Building a unified freelancer workspace that bridges client intake, planning, proposals, and delivery with **AWS** and **CI** gates.
- Shipped an email intake console that triages threads, captures scope, and renders versioned proposal PDFs via a Vision Model.
- Built a **Python** evaluator→revisor **agent** loop with rubric-based **benchmarking** for measured quality improvements.

SKILLS

Languages: Java, Python, Scala, TypeScript/JavaScript, SQL, C/C++

Platforms & Tools: Apache Spark, Airflow, Kafka, Delta Lake, AWS, Spring Boot, Flask, Docker, Git, RBDBMS